

Towards System-Friendly Solid-State Circuit Breaker for Electrified Aircraft Propulsion

Dehao Qin
Clemson University
North Charleston, USA
dehaoq@g.clemson.edu

Zheyu Zhang
Rensselaer Polytechnic Institute
Troy, USA
zhangz49@rpi.edu

Shimul K Dam
University of Tennessee
Knoxville, USA
sdam@utk.edu

Ching-Hsiang Yang
University of Tennessee
Knoxville, USA
cyang30@vols.utk.edu

Zhou Dong
ABB Inc
ABB US Research Center
Raleigh, USA
zdong9@vols.utk.edu

Cheng Wan
Clemson University
North Charleston, USA
wchghust@gmail.com

Hua Bai
University of Tennessee
Knoxville, USA
hbai2@utk.edu

Fred Wang
University of Tennessee
Oak Ridge National Laboratory
Knoxville, USA
fred.wang@utk.edu

Abstract— As the enabler of the direct current (DC) power distribution system, DC solid-state circuit breaker (DC-SSCB) has been extensively studied at the breaker-level (e.g., topology, energy absorption circuit, gate driver, and control) with limited system-level considerations, which is also crucial. This paper provides an all-in-one solution with integrated circuitries as the fault detector, actuator for semiconductor's operating status regulation, and coordinated control to seamlessly transition from one mode to another (e.g., normal mode, precaution mode, i^2t mode, and interruption mode). This allows the developed SSCB to limit system fault current not exceeding short-circuit current rating (SCCR) and take different responses under different fault cases. First, two improved fault detection circuits are introduced to better serve the SSCB application. Second, four operation modes are proposed for SSCB to realize system-friendly functions. Third, by leveraging four different operation modes, a control strategy is demonstrated to take different actions based on different fault cases. Finally, the feasibility and the effectiveness of the proposed system-friendly 200V/10A SSCB is validated with experimental results.

Keywords—solid-state circuit breaker, system-friendly, electrified aircraft propulsion

I. INTRODUCTION

Since NASA invented its first DC-SSCB in 1973 (based on the online library) [1], the study on SSCB mainly focuses on power semiconductor selection [2], topology [3], energy absorption circuit [4, 5], and control design [6], fault detection design, etc. Meanwhile, as an active participant in the distribution system's reliable operation, beyond the primary fault interruption duty, SSCB should be capable of other system-friendly functions to benefit the system operation [7]. Fig. 1 shows the basic topology and location of SSCB in an electrified aircraft propulsion system [8].

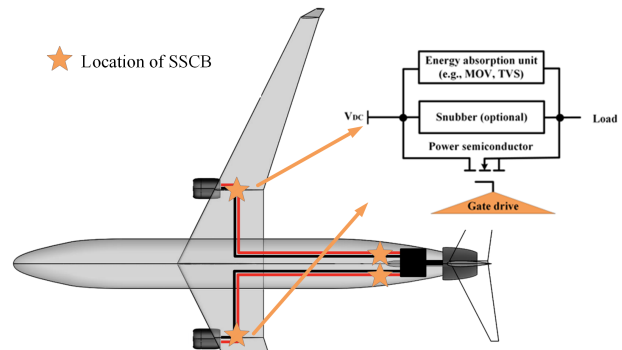


Fig. 1. SSCB in electrified aircraft propulsion.

A few studies have explored system-friendly functions for DC-SSCB, including soft start-up, fault location detection, fault current limiting, and soft reclosing.

As the system protection equipment, it is suitable for DC-SSCB to have the capability of fault location detection. The first DC-SSCB with a short-circuit (SC) fault location detection function is proposed in [9]. The proposed DC-SSCB has a unique PWM current limiting mode. In the PWM current limiting, an algorithm is leveraged to calculate the SC distance using the power line's per-unit inductance value by measuring the response of the voltage pulses injected into the DC power network.

Besides the fault location detection function, the current limiting function is also essential for DC-SSCB. For example, it is preferred to enable the DC-SSCB with an inrush current limitation function during the system start-up. A feasible method is to decrease the gate voltage once the overcurrent event is detected [10-12]. The fault current can force the power semiconductor to enter the saturation region, and the V-I curve determines the system's current limitation level. To

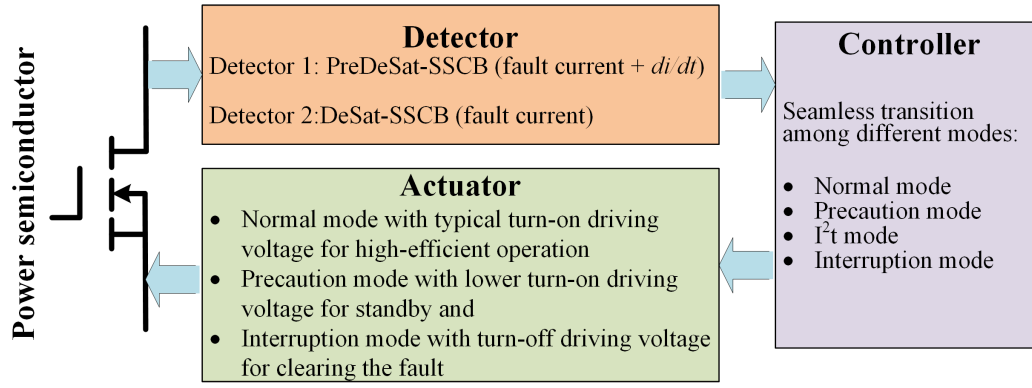


Fig. 2. System configuration of the proposed SSCB.

improve the current limiting capability of the semiconductor in the saturation region, a current limiting strategy for SSCB with series-connected switching cells in [13]. However, these studies only focus on the system's current limitation capacity of SSCB. A holistic control strategy lacks to coordinate SSCB's limiter and breaker capabilities to serve as a DC distribution system-friendly player.

It is also essential for DC-SSCB to distinguish the overcurrent and SC for DC-SSCB. Ref [14] proposes a digitally controlled current-time profile-based DC-SSCB for overcurrent protection and SC protection. The proposed current-time profile can help DC-SSCB avoid the fault triggered by the inrush current caused during the start-up of the power electronics intensive system. However, it requires an additional microcontroller to control SSCB, which is complicated and expensive.

To enable reliable operation, the electrified aircraft propulsion system is more demanding for the SSCB-based protection system:

- The fault current cannot exceed SC current rating (SCCR) defined by the system (e.g., 10X) [15].
- Limit the worst scenario fault current below SSCR and trip the SSCB instantaneously (e.g., sub- μ s).
- Interrupt the relatively low current fault, instead of instantaneous response, following the i^2t capability of SSCB for better fault type self-awareness and system protection coordination.
- Stand by and bypass a temporary fault with self-recovery capability.

The proposed system-friendly SSCB aims to provide an all-in-one solution with 1) integrated circuitries to serve as fault detectors (not only to detect the fault but also predict the fault current considering absolute fault current and di/dt) and actuator (to manage power device's operating status), and 2) coordinated control to seamlessly transition from one mode to another (normal mode, precaution mode, i^2t mode, and interruption mode). For the fault detector, by introducing the influence of Ldi/dt , an improved fault detection circuit is proposed, which considers the value and the increasing ratio of the system's current. Meanwhile, for the actuator, the precaution mode can ensure the system's current cannot be higher than SCCR; i^2t mode guides SSCB's response to the

low current fault based on the power device's thermal characteristics [16]; interruption mode is leveraged to interrupt the system's fault current. For the coordinated control, based on the different fault system scenarios, the proposed hardware-based controller can guide four different modes to enable SSCB to take action accordingly.

II. OVERVIEW OF THE PROPOSED SSCB

Fig. 2 overviews functional blocks of the proposed SSCB, including two fault detectors, three actuators, and a controller.

Based on different requirements, two fault detectors are devised: PreDeSat-SSCB and DeSat-SSCB.

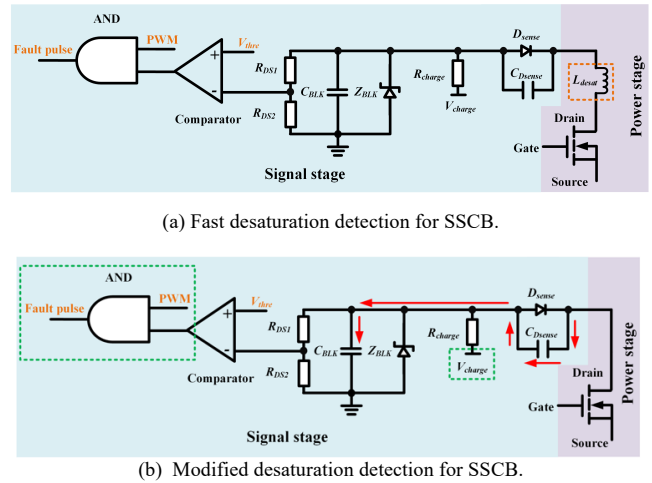


Fig. 3. Topology of the proposed fault detection circuit for SSCB.

- PreDeSat-SSCB relies on di/dt -enabled fast fault current indicator, generating the precaution signal to alert SSCB of SC risk, which is shown in Fig. 3 (a).
- DeSat-SSCB depends on a modified, SSCB-suitable desaturation detection, generating the SC tripping signal to interrupt SSCB, which is shown in Fig. 3 (b).

TABLE I. SSCB'S ACTIONS IN DIFFERENT FAULT CASES

Fault scenarios	Actions
Low impedance fault (System's fault current approaches SCCR and hits DeSat-SSCB's tripping threshold)	Step 1: Normal mode → Precaution mode Step 2: When i_{sys} hits SSCB tripping threshold, SSCB enters interruption mode to interrupt the fault current.
High impedance fault (System's fault current lower than SCCR)	Step 1: Normal mode → Precaution mode Step 2: Based on the power device's thermal performance, SSCB is interrupted by the i^2t signal from the external control unit.
Short-time overcurrent (System's start-up, fault fast recovery)	Step 1: Normal mode → Precaution mode Step 2: Back from precaution mode to normal mode, once the i_{sys} is back to normal.

Once the fault is predicted/detected by the detectors, the controller identifies the fault types and decides the actions of the actuator based on the following operation modes.

- **Normal mode:** Under normal operation, SSCB's gate voltage stays at turn-on gate voltage (V_{GS-on}) through the actuator to conduct the normal system current with high efficiency.
- **Precaution mode:** Precaution mode is triggered by the precaution signal from the di/dt-based fast fault current indicator, and SSCB's gate voltage decreases to precaution voltage (V_{prec}) through the actuator to ensure the system's current (i_{sys}) cannot exceed system's SCCR.
- **I^2t mode:** i^2t mode is triggered by an external i^2t detection unit, and SSCB's gate voltage decreases to turn-off voltage (V_{GS-off}) through the actuator to interrupt the low current fault. The interruption time depends on the power device's intrinsic thermal limit and the current i_{sys} .
- **Interruption mode:** Interruption mode is triggered by the SC tripping signal from the normal fault detector, and SSCB's gate voltage decreases to V_{GS-off} through the actuator to interrupt the current at SSCB's SC tripping threshold.

According to the inputs from the detector, the actuator will control the gate voltage to manage the operating status of the power device in SSCB.

- Under the normal mode, gate voltage remains a typical turn-on driving voltage for high-efficient operation.

- Under the precaution mode, gate voltage reduces to the level where the corresponding desaturation current follows the device's V-I curve to the SCCR.
- Under the interruption mode, gate voltage controls to turn-off driving voltage to clear the fault.

By leveraging the different operation modes, the controller can enable SSCB to take different actions according to different fault cases, as summarized in TABLE I. Once di/dt-based fast fault current indicator generates the precaution signal, SSCB enters precaution mode to ensure the maximum i_{sys} cannot exceed the system's SCCR and standby. Then, based on different fault cases, SSCB is controlled to enter different operation modes, as illustrated in Fig. 4.

III. IMPLEMENTATION AND VERIFICATION

A. Implementation

Fig. 5 shows the schematic of the proposed SSCB. The proposed SSCB is composed of detector, controller and actuator.

Detector

The detection circuit is composed of DeSat-SSCB and PreDeSat-SSCB. The sensitive PreDeSat-SSCB is leveraged to detect the potential SC fault. Meanwhile, the DeSat-SSCB is leveraged to detect SSCB's SC tripping threshold current. To make the precaution mode more sensitive to potential SC fault, the threshold voltage of PreDeSat-SSCB V_{thre1} should be relatively small. Meanwhile, the threshold voltage of DeSat-SSCB V_{thre2} needs to fit with SSCB's SC tripping threshold current.

Controller

The controller comprises several logic gate ICs to generate SSCB's control signal. The precaution pulse is generated by PreDeSat-SSCB, while DeSat-SSCB generates the interruption pulse. The i^2t signal is from the external control unit. The generated precaution pulse is sent to an AND gate (AND 3). Meanwhile, the generated interruption pulse is sent to an OR gate (OR 1) with the external i^2t signal. The output of OR 1 will be sent to a latch-up circuit to generate the turn-off signal to interrupt the system's current. The precaution signal will AND with the inversed turn-off signal generating the

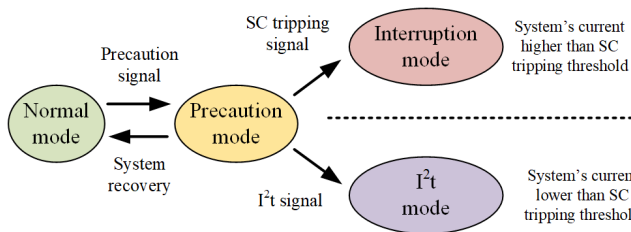


Fig. 4. Operation modes transformation.

TABLE II. DETAIL OF DIFFERENT OPERATION MODES

Mode	Detector			Actuator		
	DeSat	PreDeSat	Ext. i^2t	V_{GS-on}	V_{prec}	V_{GS-off}
Normal	×	×	×	○	×	×
Precaution	×	○	×	×	○	×
I^2t	×	×	○	×	×	○
Interruption	○	×	×	×	×	○

○: with ×: without

B. Verification

A GaN-based 200V/10A SSCB is prototyped to demonstrate the feasibility and effectiveness of the proposed system-friendly SSCB, as shown in Fig. 8. TABLE III summarizes the key parameters. The test platform is mainly composed of the DC power supply VDC, fault trigger breaker, system fault impedance Z_{fault} , and the proposed SSCB. A separate fault trigger breaker is introduced to create the SC fault events. Once the fault trigger breaker is conducted, it can be assumed that the SC occurs. Z_{fault} can emulate the fault system impedance once the SC occurs. The response of the proposed SSCB to different SC fault cases could be tested by adjusting Z_{fault} .

TABLE III. PARAMETERS OF THE SSCB PROTOTYPE

Parameters	Value
Power device	GS66516T
Snubber	150nF, 0.5Ω
TVS	AK3-430
SCCR	150A
DC bus voltage	200V
Turn-on voltage	6V
Turn-off voltage	-3V
Precaution voltage	4V
Gate Drive IC	IXDD609

Low impedance fault

Fig. 9 illustrates the waveform of the SSCB when the fault system inductance L_{fault} is 0.7 μH, which is primarily contributed by interconnecting wire instead of a dedicated inductor. The SC fault happens at 0 μs, and v_{GS} is 6 V. Then, at approximately 0.5 μs, the PreDeSat-SSCB detects the potential SC, SSCB enters precaution mode, and v_{GS} drops to 4 V, which corresponds to the predefined SCCR in TABLE III. DeSat-SSCB detects the overcurrent at about 1.4 μs, SSCB enters interruption mode, and v_{GS} drops from 4 V to -3V to interrupt the system's current. Based on the waveform, it can be found that the proposed SSCB can interrupt a low impedance fault in 1 μs; meanwhile, the current flowing through the power device i_D is limited by the precaution voltage to 140 A stably.

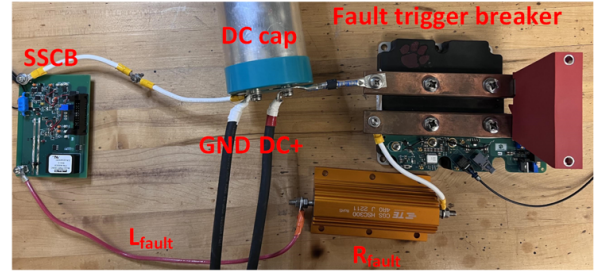
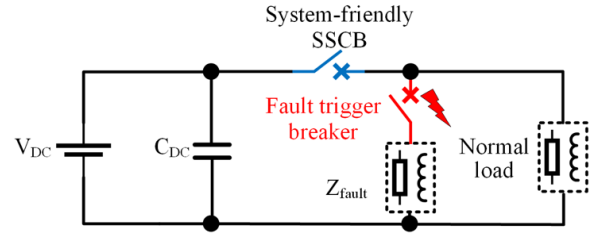
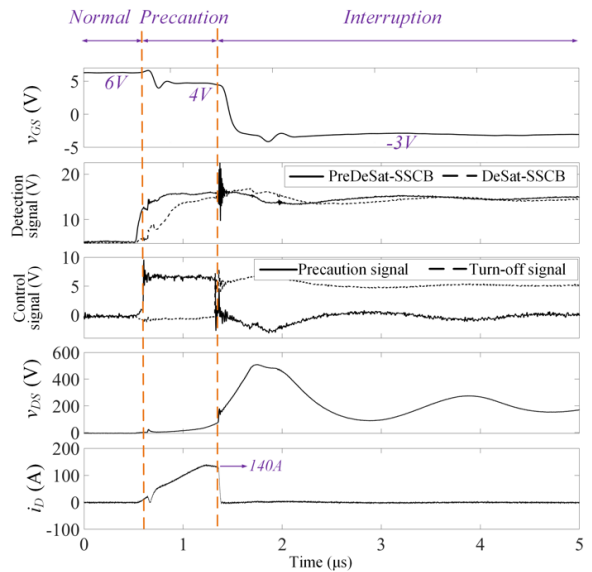


Fig .8. Test platform for the proposed SSCB.

High impedance fault

Fig. 10 illustrates the waveform of the SSCB with 2 μH system inductance L_{fault} and 4 Ω system resistance R_{fault} . The system's maximum fault current is limited to 50A by R_{fault} . The SC fault happens at 0 μs, and v_{GS} is 6 V. At 0.8 μs, the PreDeSat-SSCB detects the potential SC, SSCB enters precaution mode, and v_{GS} drops to 4 V. Approximately 4 μs later, SSCB receives an emulated i^2t signal for the sake of test and enters i^2t mode. I^2t signal generates a turn-off signal and v_{GS} drops from 4V to -3V to interrupt the system's fault current. Because the v_{DS} does not achieve the threshold voltage for DeSat-SSCB, DeSat-SSCB is not triggered in this case. The test results agree with the analysis above.

Fig. 9. Test waveform under the low impedance fault ($L_{fault}=0.7\mu H$).

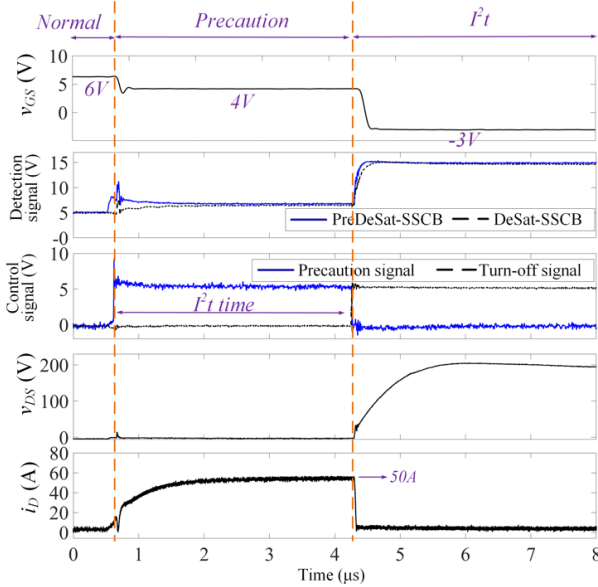


Fig. 10. Test waveform under the high impedance fault ($L_{fault}=2 \mu H$, $R_{fault}=4 \Omega$).

Fault fast recovery case

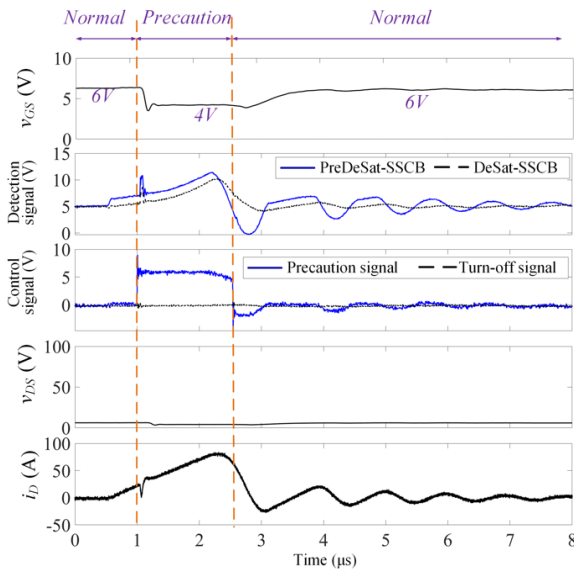


Fig. 11. Test waveform under the fault fast recovery case.

Fig. 11. illustrates the waveform of the SSCB under a fault fast recovery case. The SC fault happens at 0 μs with 6 V v_{GS} . Then, at 1 μs , the PreDeSat-SSCB detects the potential SC, and SSCB enters precaution mode with 4 V v_{GS} . Then, the fault recovers at 2.2 μs . With the fault recovery, the system's current decreases. Accordingly, PreDeSat-SSCB detected v_{DS} decreases and becomes lower than PreDeSat-SSCB's threshold voltage. Then, the precaution signal disappears, and

v_{GS} increases from 4V to 6V to resume the system's normal operation. During this process, because the fault time is shorter than i^2t time and the system's fault current is not high enough, i^2t and interruption modes are not triggered.

IV. CONCLUSION

A system-friendly SSCB is developed and demonstrated. The key takeaways and findings are summarized below.

- 1) Two fault current detection circuits, DeSat-SSCB and PreDesat-SSCB, are suitable for the SSCB application. Particularly, the PreDeSat-SSCB, considering both the fault current absolute value and di/dt , results in the increased fault detection response speedup by 10 X as compared to conventional desaturation detection based on the test.
- 2) Four operation modes, including normal mode, precaution mode, i^2t mode, and interruption mode, are offered: i) normal mode with the normal driving voltage makes the efficient operation of the SSCB, ii) precaution mode with reduced driving voltage ensures the system's fault current lower than SCCR; iii) i^2t mode helps SSCB interrupt low fault current considering power devices' intrinsic thermal limit; iv) interruption mode allows SSCB's interruption at SCCR in 100s of ns to 1 μs response time.
- 3) Coordinated control enables SSCB to take corresponding actions to various fault cases: i) once the SC and/or high di/dt event is detected through PreDeSat-SSCB, the proposed SSCB lowers the power device gate voltage to ensure the system's maximum fault current below SCCR; ii) under the low-impedance fault when the fault current approaches SCCR, the proposed SSCB based on DeSat-SSCB triggering signal can interrupt the fault within 200 ns to 1 μs ; iii) under the high-impedance fault when the fault current does not exceed SCCR, the proposed SSCB can interrupt the fault based on an external i^2t detection and protection unit; iv) finally, the SSCB is capable of fast fault recovery, i.e., when the fault is cleared by a local SSCB and the protection signal by PreDesat-SSCB disappears, the SSCB is able to be back under the normal operation.

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